

Amir Naseredini

Senior CPU Engineer at Huawei Technologies R&D, UK
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Interests

- Open Source Software
- Rowhammer Attack
- Vulnerability Analysis
- Secure Information Flow
- Programming Languages
- Architecture & Microarchitecture
- Microarchitectural Attacks
- Vulnerability Patching
- Formal Security
- Linux Kernel

Experience

- Main.....
- Huawei Technologies R&D** **UK**
Senior CPU Engineer *July 2024–Current*
- To analyse product architectures and microarchitectures in order to design solutions for optimisations
- Canonical** **London, UK**
Security Engineer *Jan. 2023–June 2024*
- To analyse, fix, and test vulnerabilities in Ubuntu packages
 - To keep track of vulnerabilities in Ubuntu releases as they are discovered, researched, and fixed
 - To maintain Node.js security in Ubuntu
 - To review snaps before granting non-default privileges to them at the Snap Store
 - To audit source code for vulnerabilities
- Google** **London, UK**
Security Engineer Intern *Sep. 2022–Dec. 2022*
- To develop kernel modules and device drivers
 - To analyse VirtIO devices
 - To develop a full stack device in crosvm in order to make the DRAM analysis easier
- Royal Holloway, University of London** **London, UK**
Post-Doctoral Research Assistant *Mar. 2022–Sep. 2022*
- To carry out research on Active Automata Learning and DRAM security
 - To develop our open source tool, ALARM, to analyse a DRAM model against Rowhammer
- Amirkabir University of Technology (Tehran Polytechnic)** **Tehran, Iran**
Security Researcher *Feb. 2017–Sep. 2018*
- To carry out research on computer security assessment
 - To perform web application penetration testing
- Miscellaneous.....
- University of Sussex** **Brighton, UK**
Associate Tutor *Sep. 2018–Sep.2022*
- To help running various modules in the Department of Informatics

TU Graz*Visiting Researcher***Graz, Austria***Sep. 2020–Mar. 2021*

- To carry out research on Microarchitectural Attacks and Programming Languages and Execution Environments
- To develop a tool, Speconnector, to analyse and perform Spectre independent of the target language and execution environment

Amirkabir University of Technology (Tehran Polytechnic)*Teaching Assistant***Tehran, Iran***Sep. 2015–Jan.2018*

To help running various modules in the Department of Computer Engineering

University of Kurdistan*Teaching Assistant***Sanandaj, Iran***Sep. 2012–May.2015*

To help running various modules in the Department of Computer Engineering

Education

University of Sussex*PhD, Informatics (Computer Science)***Brighton, UK***Sep. 2018–Nov. 2023*

Thesis title: "Towards Automatic Analysis of Microarchitectural Attacks"

Amirkabir University of Technology (Tehran Polytechnic)*MSc, Information Security Eng.***Tehran, Iran***Sep. 2015–Feb.2018*

Thesis title: "Algebraic Cryptanalysis of ARX-Design Hash Functions"

University of Kurdistan*BSc, Information Technology Eng.***Sanandaj, Iran***Sep. 2011–July 2015*

Received the 1st Student Award, in the Computer Engineering and Information Technology Department with overall GPA of 92.35%

Awards, Certificates, Publications, and Presentations

Awards*Notable Awards*

- Awarded with the Star of Cambridge by Huawei Technologies Research & Development UK for high impact deliveries, 2025
- Awarded with the School of Engineering and Informatics' Fully-Funded Scholarship by the University of Sussex, 018
- Ranked #2 in the MSc National Entrance Exam in Iran (Having my BSc GPA Considered), 2015
- Selected as a Talented Student, Two Times in a Row at the University of Kurdistan, 2013 to 2015

Certificates*Linux Foundation*

- Developing Secure Software (LFD121), 2024
- Security and the Linux Kernel (LFD441), 2024
- Linux Kernel Internals and Development (LFD420), 2023
- A Beginner's Guide to Linux Kernel Development (LFD103), 2023
- Open Source Licensing Basics for Software Developers (LFC191), 2023

Publications

Notable Projects

- A. Naseredini, M. Berger, M. Sammartino, S. Xiong, "ALARM: Active LeArning of Rowhammer Mitigations", Hardware and Architectural Support for Security and Privacy (HASP) 2022, October 1, 2022 – co-located with MICRO 2022
- A. Naseredini, S. Gast, M. Schwarzl, P. Bernardo, A. Smajic, C. Canella, M. Berger, D. Gruss, "Systematic Analysis of Programming Languages and Their Execution Environments for Spectre Attacks", 8th International Conference on Information Systems Security and Privacy (ICISSP2022), February 2022

Presentations

Notable Projects

- "ALARM: Active LeArning of Rowhammer Mitigations". Presented at the Informatics department, King's College London, UK, March 2023
- "Systematic Analysis of Programming Languages and Their Execution Environments for Spectre Attacks". Presented at the Computer Science department, University College London (UCL), UK, February 2022

Skills

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|----------------------|-------------------|
| ○ Debian based Linux | ○ Git |
| ○ C/C++ | ○ Python |
| ○ Rust | ○ Java |
| ○ Haskell | ○ Bash |
| ○ Perfetto | ○ Simpleperf |
| ○ Jira | ○ Coverity |
| ○ Metasploit | ○ Wireshark |
| ○ Nmap | ○ Tenable Nessus |
| ○ LaTeX | ○ Public Speaking |

Membership and Services

- Student Volunteer at ECOOP and Curry-On 2019
- Student Volunteer at PLDI 2020

Languages

Kurdish: Native

Persian: Native

English: Full Professional

Arabic: Elementary

Hobbies

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| ○ Running | ○ Listening to Music |
| ○ Walking | ○ Reading |
| ○ Rubik Solving | ○ Sudoku Solving |
| ○ Physical Fitness | ○ Volleyball |
| ○ Swimming | ○ Football |